Temporal Visual Intelligence: Disentangling Sequential Order from Visual Evidence Accumulation

A Controlled Investigation into Sequential Dynamics, Representation Trajectories, and Multimodal Semantic Grounding

Document: Research project report
Evaluation Benchmark: 20-class directional diagnostic dataset (source identity/provenance unresolved; manifest labels it Something-Something V2) (N = 2,000 sequences, 20 classes, N=300 held-out test split)
Status: Research report; selected checkpoint diagnostics verified, data provenance and secondary claims unresolved
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Abstract

This report describes a checkpoint-based comparison of single-frame, order-invariant pooling, and recurrent classifiers on a 20-class, 2,000-sequence directional diagnostic dataset. Dataset identity remains unresolved: the manifest names Something-Something V2, while configuration and preparation code also describe synthetic data. We cannot confirm the source or license of the tracked images.

The local checkpoint diagnostics show Model C at 34.78% accuracy and Model D at 23.56%, compared with 8.78% for Model A and 8.22% for Model B. Five paired comparisons (H1-H4 and H14) were recomputed from saved prediction rows. Their uncertainty estimates condition on saved checkpoints, average correctness differences over seeds at the sequence level, and omit training-seed variability; this is not independent retraining. Other M6/M5 analyses and cohort claims remain unverified. The annotations are heuristic, not a live VLM or independent human validation. A clean Git clone lacks the local checkpoints and feature arrays required to reproduce these values.

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1. Problem Formulation & Hypotheses

1.1 Core Research Question

The assessment investigates: \text{Can a model extract and utilize useful information from a sequence of visual observations that may not be available from an individual frame?}

In video classification, high performance is frequently misattributed to complex temporal reasoning when it is actually driven by static background scene correlations, object presence, or multi-view evidence accumulation. We formulate three competing operational mechanisms: 1. **Mechanism 1 (Visual Evidence Accumulation)**: Additional observations increase the statistical likelihood of observing discriminative visual features, solvable via symmetric order-blind pooling. 2. **Mechanism 2 (Chronological Order Sensitivity)**: Directionally paired transformations where discriminative information resides exclusively in sequence order (t1, \dots, tK). 3. **Mechanism 3 (Contextual Error Recovery)**: Recurrent temporal state consistency compensating for frame corruption, occlusion, or sensor noise.

1.2 Assessment-Aligned Research Questions and Experimental Predictions

The assessment asks for an open-ended investigation, not preregistered hypotheses. The following predictions were used to organize experiments; their status must be determined from reconciled outputs: - **P1 (Progressive observations)**: Compare performance across increasing observation horizons. - **P2 (Order sensitivity)**: Compare canonical and shuffled sequences while preserving frames. - **P3 (Corruption)**: Compare models under missing or corrupted observations. - **P4 (Time reversal)**: Test whether reversing a sequence changes predictions. - **P5 (Representation sampling)**: Inspect representation trajectories under different frame sampling. A Nyquist bound is not claimed without validated physical frequency measurements.

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2. Dataset Curation, Representation & Experimental Parity

2.1 Two-Tier Benchmark Architecture

1. **Tier 1 (Directional diagnostic dataset)**: 2,000 sequences across 20 classes according to local metadata. Source identity is unresolved: the manifest labels the set Something-Something V2, while config/preparation code includes synthetic data modes. We cannot verify that these are SSv2 samples or rule out dataset shortcuts.
2. **Tier 2 (Continuous clip inspection)**: A small set of continuous clips is used for representation and trajectory inspection. It is not a labeled real-world action classification benchmark and does not establish quantitative transfer performance.
3. **Group-Aware Split Check**: Stored split metadata records 1,400 train, 300 validation, and 300 test sequence IDs with no ID overlap. Broader physical-event or identity leakage has not been independently audited for this release.

2.2 Frozen Foundational Vision Encoders

1. **Primary Backbone**: DINOv2 ViT-S/14 (384-dimensional latent embedding) evaluated in strict eval() mode ($\nabla_\theta L = 0$).
1. **Secondary Backbone (M6)**: OpenCLIP ViT-B/32 artifacts exist, but paired predictions, preprocessing, checkpoints, and source provenance are not verified; no cross-encoder finding is claimed.

2.3 Architectural Specifications & Capacity Parity Triad

The model designs include exact parameter parity controls; checkpoint diagnostics alone do not establish that all training/evaluation conditions were otherwise matched:

Architecture	Input Formulation	Parameters	Chronological Order Sensitivity
Model A (SingleFrameMLP)	Single observation e_t (middle frame)	**103,700**	None (Single frame anchor)
Model B (MultiFrameMeanPool)	Symmetric average {e} = (1)/(K)_{t=1}^K e_t	**103,700**	Permutation Invariant (_{shuffle} = 0.00 pp)
Model C (TemporalGRU)	Sequence e_{1:K} (Projection 256, Hidden 128)	249,364	Recurrent state ht = GRU(h{t-1}, z_t)
Model D (CompactTemporalGRU)	Sequence e_{1:K} (Projection 130, Hidden 64)	**103,700**	**Exact Capacity Parity with Models A & B**

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3. Main Quantitative Findings and Diagnostic Checks

3.1 Headline Model Comparison (Checkpoint Diagnostic)

Current checkpoint recomputation (Seeds 42, 43, 44) on the N=300 test partition yielded the following diagnostic means. These are not independent retraining results. The saved predictions support five recomputed paired contrasts (H1–H4 and H14; see the statistical table); inference is conditional on these checkpoints and does not capture training-seed variability. Other historical inferential statistics remain unverified.

Model Architecture	Parameter Count	Accuracy (Mean ± Population SD)	Macro-F1 (Mean ± Std)	Paired comparison vs Model B (95% CI, pp)
Model A: Single-Frame MLP	103,700	8.78 \pm 3.28\%	0.0327 \pm 0.0251	Reference for H1; not vs Model B
Model B: MultiFrame MeanPool	103,700	8.22 \pm 1.26\%	0.0216 \pm 0.0090	Diagnostic parity reference
Model D: Compact Temporal GRU	**103,700**	**23.56 \pm 2.20\%**	**0.1800 \pm 0.0339**	+15.33 pp [12.33, 18.33] (H14)
Model C: Standard Temporal GRU	249,364	34.78 \pm 1.10\%	0.3204 \pm 0.0057	+26.56 pp [22.11, 31.11] (H2)

Diagnostic checkpoint comparison: These means and population standard deviations were recomputed from the current checkpoints on the test split. The paired table reports sequence-level bootstrap intervals and Monte Carlo sign-flip tests for its five verified comparisons, with Benjamini–Hochberg correction across that five-test family. All five observed permutation samples had zero exceedances; the p-values are bounded by the 100,000-permutation resolution, not exact zero probabilities. This is not an independent rerun of training. The M4 summary file lists a different CompactGRU mean; its source protocol remains unresolved.

3.2 Progressive Horizon Sweep (Descriptive Accuracy)

The progressive prediction CSV gives Model C accuracy: K=1 11.56%, K=2 14.22%, K=4 22.67%, and K=8 34.78% (three-seed means). These are descriptive checkpoint diagnostics. The sequence-level H1 contrast is K=8 GRU versus single-frame baseline; no inferential horizon-scaling claim is made.

3.3 Paired Order Comparisons

The source prediction records compare model outputs under saved order conditions. These comparisons are conditional on saved checkpoints and do not establish a causal mechanism: - **Temporal Shuffle**: The saved per-sequence predictions and corrected checkpoint recomputation give 34.78% canonical and 20.67% shuffled accuracy (14.11 pp difference). A sequence-level paired bootstrap gives a 95% interval of [10.78, 17.56] pp; the Monte Carlo sign-flip test found zero exceedances in 100,000 permutations. Historical M4/M6 summaries report 13.89 pp and use unverified inferential values. - **Temporal Reversal**: The saved per-sequence predictions give an 18.89 pp accuracy drop (95% paired bootstrap interval [15.11, 22.89] pp); the sign-flip test found zero exceedances in 100,000 permutations. The M6 reporter's historical 4.67 pp H4 contrast is inconsistent with these inputs.

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4. Semantic Failure Analysis, Taxonomy & VLM Grounding

4.1 Stored Cohort Tables (Unverified)

M4/M5 cohort tables disagree in counts and denominators. Their prevalence and class-specific narratives are not included as verified results; recompute from a single canonical per-sequence prediction artifact before citing them.

4.2 Information Attribution (Not Established)

Aggregate differences in accuracy do not identify separate causal information components. Historical cohort-level attribution conflicts with the aggregate measurements and is not treated as a result here.

4.3 Offline Taxonomy Annotations and Annotation Limitations

The saved 127-sequence annotations were generated by offline_kinematic_engine, a deterministic heuristic using pixels, embeddings, and class-taxonomy priors. They are not outputs from a VLM. The accompanying verifier starts each “human” record by copying the candidate annotation and then applies deterministic rules; therefore the stored 97.15% field agreement and Cohen's kappa do not establish independent human inter-rater reliability. Taxonomy prevalence and cohort claims should be treated as heuristic outputs until independently blinded human annotations are collected and analyzed. The code includes a live Gemini backend, but no live VLM output is represented in the saved annotations.

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5. Generalization, Robustness & Shortcut Controls (M6)

The repository contains M6 artifacts for additional shifts, but their paired prediction inputs and protocols have not been reconciled. Their stored values are not presented as verified findings here: 1. **Sampling Strategy Robustness**: Uniform, sparse, dense, random, and jittered sampling summaries are stored; per-sequence inputs and inferential results were not verified in this audit. 2. **Missing Observations**: Frame-drop and corruption curves are stored, but the current report does not treat their historical accuracy or graceful-degradation claims as recomputed evidence. 3. **Cross-Encoder Verification**: A DINOv2/CLIP summary is present. The encoder-specific predictions, checkpoints, and preprocessing provenance still require verification. 4. **Shortcut Probes & Negative Controls**: Implementations and summary outputs exist. Their results may constrain specific shortcut explanations only after their frozen inputs and controls are validated; they cannot establish the absence of all dataset artifacts. 5. **Statistical Significance**: Five contrasts with complete per-sequence artifacts (H1–H4 and H14) were recomputed and survive Benjamini–Hochberg adjustment within that five-test family; intervals and tests condition on saved checkpoints. The remaining M6 comparisons are not verified because their complete paired prediction inputs were not available. See methods and results.

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6. Real-World Manifold Transfer & Limitations

6.1 Photographic Real-World Validation

Inspecting continuous vehicle drifting (BMW_M4) and quadruped locomotion (Horse_running) clips produced representation-level trajectory observations. These clips are not a labeled classification test set; the observations below are descriptive and must not be presented as evidence of real-world classification transfer: - High frame-to-frame continuity in the selected clips, as shown in the associated representation plots. - Cumulative representational drift in the selected examples. - A reversal comparison in an experimental artifact; its relation to the synthetic benchmark’s reversal score should be documented separately.

6.2 Limitations & Technically Honest Disclosures

1. **Capacity Parity Comparison**: Compact Model D achieves 23.56\% accuracy at 103,700 parameters compared to 34.78\% for larger Model C (249,364 parameters). This checkpoint comparison is consistent with an order-aware model advantage over mean pooling, but does not establish a general mechanism; it is conditional on the saved checkpoints and unresolved dataset provenance.
2. **Camera Ego-Motion Confound**: In single-camera setups, rapid camera pan can mimic object translation (T_1), inducing occasional Case 3 regressions.
3. **Discrete Frame Sampling**: Uniform 8-frame sampling introduces temporal discretization. No measured frequency bound establishes a required capture rate; rates such as >60 FPS are design hypotheses, not results.

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7. Multimodal Scaling & Future Directions

We provide a conceptual system design specification (docs/multimodal_system_design.md) detailing a proposed two-speed system. Architecture, throughput, and latency figures are design targets, not measured results from this repository: - **Fast Kinematic Control Loop (30–100 Hz; design target)**: Fuse RGB, depth, 6-DoF IMU, and pose through causal cross-attention into a multi-scale recurrent core (target latency < 5 ms). These rates and latency are proposed requirements, not measured results. - **Slow Semantic VLM Loop (1–5 Hz; design target)**: An asynchronous vision-language model could provide out-of-band state-transition explanations and contextual goal priors. This rate is a proposed target; no such integrated system was evaluated here.

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8. Conclusion

The current artifacts motivate a controlled comparison of order-aware sequence models with single-frame and order-invariant pooling controls. Conflicting metric provenance and non-independent semantic annotation artifacts remain open issues. The conclusions should be narrowed to results that can be reproduced from frozen checkpoints and paired predictions; no universal claim about temporal reasoning follows from this benchmark alone.